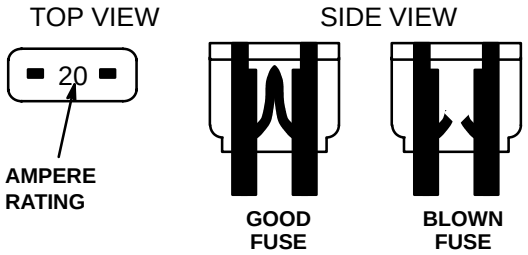
11-1 FUSE PANEL/CIRCUIT PROTECTION

CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fusible links, maxi-fuses, circuit breakers, or a combination of these devices.

BLADE TYPE FUSE

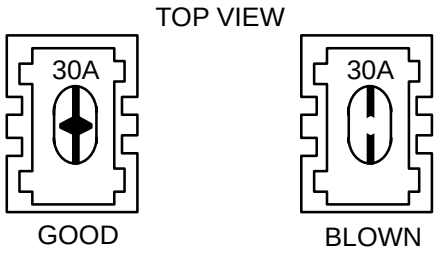


Blade type fuses have a transparent plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code shown here:

BLADE FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
4	Pink
5	Tan
10	Red
15	Blue
20	Yellow
25	Natural
30	Green

MAXI-FUSE



Cartridge maxi-fuses have transparent plastic tops. To check a cartridge maxi-fuse, look at the fuse element through the top of the housing.

To replace a cartridge maxi-fuse, pull it from the fuse box or panel. Always replace a blown cartridge maxi-fuse with a new one having the same ampere rating.

The ampere rating of a cartridge maxi-fuse can also be determined by following the color code shown here:

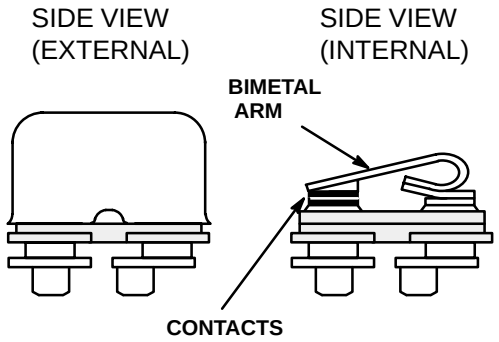
CARTRIDGE TYPE FUSIBLE LINKS	
High Current Fuse Value Amps	**Color Code**
20A Cartridge	Blue
30A Cartridge	Pink
40A Cartridge	Green
50A Cartridge	Red

CIRCUIT BREAKER

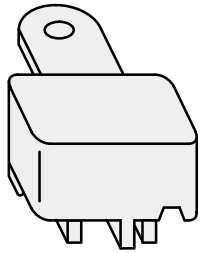
Some circuits are protected by circuit breakers (abbreviated “c. b.” in fuse chart). They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other the arm bends, opening the contacts. Current flow is broken. A circuit breaker can be the cycling or non-cycling type.

FUSE PANEL MOUNTED CYCLING TYPE

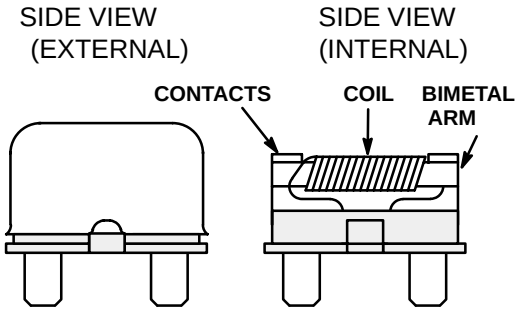


IN-LINE MOUNTED CYCLING TYPE

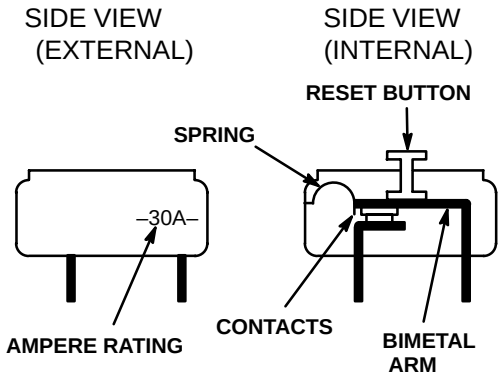


In the cycling type, the bimetal arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

FUSE PANEL MOUNTED NON-CYCLING TYPE



FUSE PANEL MOUNTED MANUAL RESET TYPE

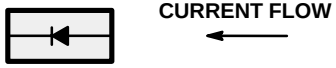


Two types of non-cycling circuit breakers are used; one is reset by removing power from the circuit, and the other is reset by depressing a reset button.

In the first type, there is a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

In the second type, a spring pushes the bimetal arm down and holds the contacts together. When an overcurrent condition exists and the bimetal arm heats up, the bimetal arm bends enough to overcome the spring and the contacts snap open. The contacts stay open until the reset button is pushed and the contacts snap together again.

DIODE



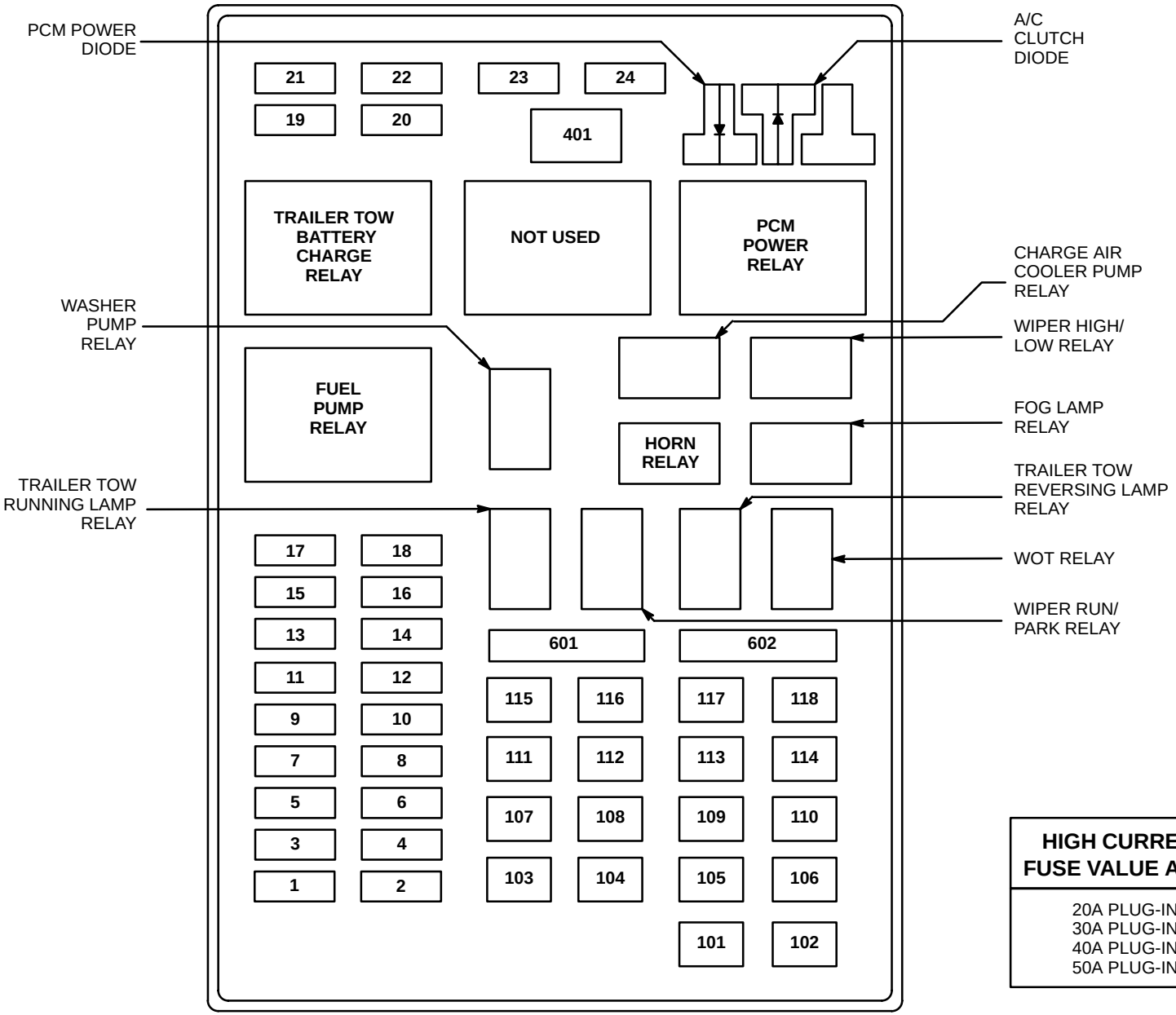
Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

MEGA FUSE

TOP VIEW

Mega fuse protects the charging circuit. In the event that the mega fuse is blown due to the charging circuit failure, the generator field circuit is disabled.

BATTERY JUNCTION BOX

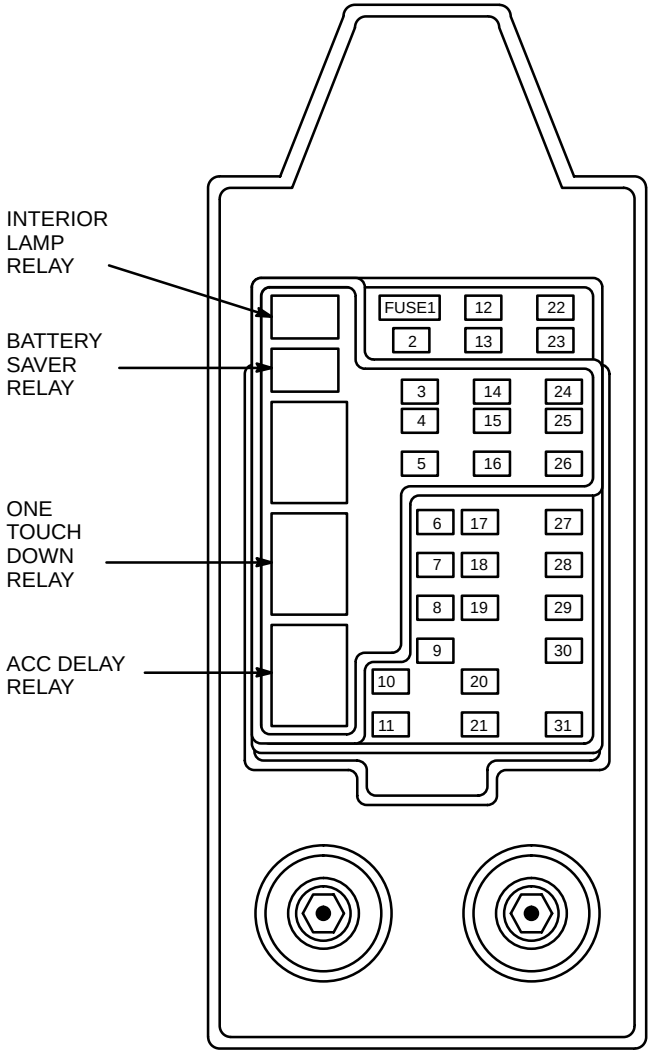


HIGH CURRENT FUSE VALUE AMPS	COLOR CODE
20A PLUG-IN	YELLOW
30A PLUG-IN	GREEN
40A PLUG-IN	ORANGE
50A PLUG-IN	RED

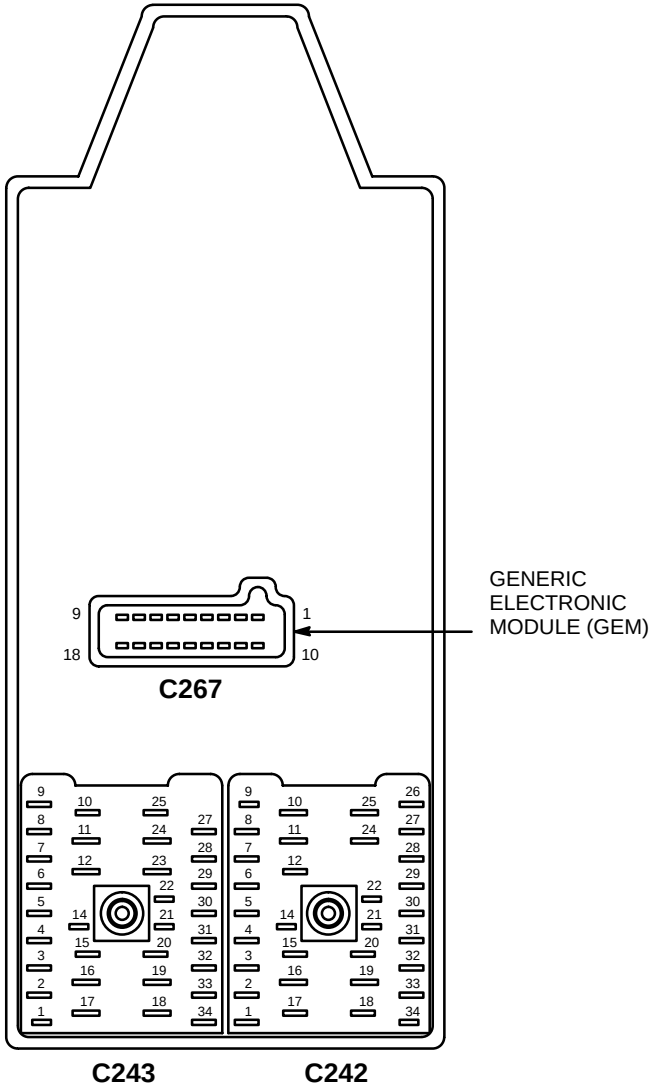
Mini-Fuses	Amps	Circuits Protected
1	20	Auxiliary Power Socket
2	30	PCM Power Relay
3	30	Main Light Switch, Headlamp Relay, Multifunction Switch
4	15	NOT USED
5	20	Trailer Tow Running Lamp Relay, Trailer Tow Reversing Lamp Relay
6	15	Main Light Switch, Park Lamp Relay
7	20	Horn Relay
8	15	Power Door Locks, CSM, Lock Relays
9	15	Daytime Running Lamps (DRL) Module, Fog Lamp Relay
10	20	Fuel Pump Relay
11	20	Charging System
12	20	Rear Power Point (Supercrew)
13	15	WOT Relay
14	–	NOT USED
15	–	NOT USED
16	–	NOT USED
17	–	NOT USED
18	15	Fuel Injectors, PCM, Fuel Pump Relay, MAF Sensor
19	10	Trailer/Camper Adapter (Right Stop and Right Turn Lamp)
20	10	Trailer/Camper Adapter (Left Stop and Left Turn Lamp)
21	–	NOT USED
22	–	NOT USED
23	15	Heated Oxygen Sensors, Transmission
24	–	NOT USED

Maxi-Fuses	Amps	Circuits Protected
101	30	Trailer Tow Battery Charge Relay
102	50	4WABS Module
102	20	Ignition Switch, RABS Module
103	50	Central Junction Box
104	30	Transfer Case Shift Relay Module, Transfer Case Electric Clutch Relay (Electronic)
105	40	Blower Relay
106	20	Charge Air Cooler Pump Relay (5.4L Sport)
107	–	NOT USED
108	30	Trailer Electronic Brake Control
109	–	NOT USED
110	30	ACC Delay Relay
111	40	Ignition Switch
112	30	Power Seat Control Switch
113	40	Ignition Switch
114	–	NOT USED
115	20	Power Door Locks, CSM (Supercrew), Lock Relays (Supercrew)
116	–	NOT USED
117	–	NOT USED
118	30	Heated Seats
601 C.B.	30	Acc Delay Relay (Supercrew)
602 C.B.	–	NOT USED

CENTRAL JUNCTION BOX



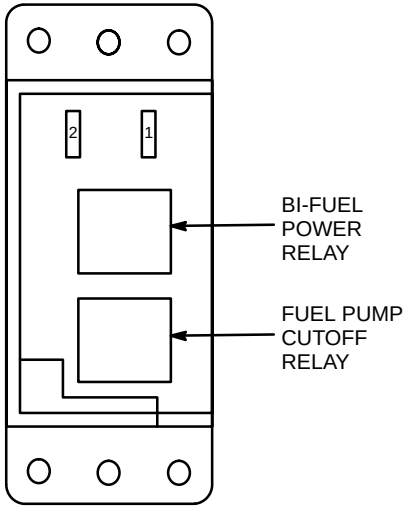
TOP VIEW



BOTTOM VIEW

Fuse Position	Amps	Circuits Protected
1	15	Radio, CD Changer, Rear Seat Entertainment
2	5	Instrument Cluster, Powertrain Control Module (PCM)
3	20	Cigar Lighter, Data Link Connector (DLC), Remote Anti-theft Personality (RAP) Module
4	5	Exterior Rear View Mirror Switch, Mirror Turn Signal Relays
5	15	Digital Transmission Range (DTR) Sensor, Speed Control Servo, Blend Door Actuator, Function Selector Switch, Daytime Running Lamps (DRL) Relay Block, Reversing Lamps
6	5	Shift Lock Actuator, Generic Electronic Module (GEM), Instrument Cluster
7	–	NOT USED
8	5	Radio, GEM, Remote Anti-theft Personality (RAP) Module, Rear Seat Entertainment
9	–	NOT USED
10	–	NOT USED
11	30	Washer Pump Relay, Wiper Run/Park Relay, Wiper HIGH/LOW Relay, Windshield Wiper Motor
12	–	NOT USED
13	20	Brake Pedal Position (BPP) Switch, Brake Pressure Switch, Flasher Relay
14	15	Battery Saver Relay, Interior Lamp Relay
15	5	Generic Electronic Module (GEM), Brake Pedal Position (BPP) Switch, RABS Module
16	20	Instrument Cluster, High Beam Headlamps (Power supplied through Multifunction Switch)
17	–	NOT USED
18	5	Main Light Switch
19	–	NOT USED
20	5	Powertrain Control Module (PCM), Generic Electronic Module (GEM), Radio
21	15	Clutch Pedal Position (CPP) Switch, Digital Transmission Range Sensor
22	10	Restraints Control Module, Passive Deactivation (PAD) Module
23	10	Trailer Tow Battery Charge Relay, Flasher Relay, 4X4 and 4X2 Center Axle Disconnect Solenoid, Transfer Case Electric Clutch Relay (Electronic), Shift on the Fly Relay (Mechanical), 4 Wheel Anti-Lock Brake System (4WABS) Module, Heated Seats, Electrochromatic Mirror, Overhead Console
24	10	Function Selector Switch Assembly
25	–	NOT USED
26	10A	Right Low Beam
27	5	Main Light Switch, Fog Lamp Relay
28	10	Left Headlamp
29	5	Autolamp Module, Transmission Control Switch (TCS), Central Security Module
30	30	Radio Noise Capacitor, Ignition Coils, PCM Power Diode, Coil on Plugs, Instrument Cluster, Anti-Theft Transceiver Module
31	–	NOT USED

BI-FUEL RELAY MODULE



Fuse Position	Amps	Circuits Protected
1	3	Compuvalve Module
2	10	Bi-Fuel Power Relay, Low Flow Injector Assembly, High Flow Injector Assembly